

JISU NAM

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RESEARCH INTEREST

My research bridges visual perception and generation. I am interested in general perception for images and videos, particularly visual correspondence, such as image matching and point tracking. Recently, I have been working on large-scale image/video generative models, analyzing their representations in the context of correspondence, ultimately to understand, enhance, and manipulate the synthesis process.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) M.S./Ph.D. in Artificial Intelligence Supervisor: Seungryong Kim	Sep. 2024 - Present <i>Seoul, Korea</i>
Korea University M.S./Ph.D. in Computer Science and Engineering Supervisor: Seungryong Kim	Sep. 2022 - Aug. 2024 (Incompleted) <i>Seoul, Korea</i>
Korea University B.S. in Biomedical Engineering and Artificial Intelligence (Double Majors)	Mar. 2018 - Aug. 2022 <i>Seoul, Korea</i>

EXPERIENCE

Meta Reality Labs Research Intern Mentors: Zhicheng Yan, Xiaoyu Xiang	Jun. 2026 - Nov. 2026 (Expected) <i>Burlingame, California</i>
Adobe Research Research Intern Mentors: Yang Zhou, Yicong Hong, Chun-Hao Huang, Feng Liu Project: Foundation Model for Interactive Autoregressive 3D Gaming Worlds	May. 2025 - Sep. 2025 <i>San Jose, California</i>
Google DeepMind Research Collaboration Mentors: Junhwa Hur Project: Video Diffusion Transformer for Temporal Correspondence [C9], [P3]	Dec. 2024 - Dec. 2025 <i>Cambridge, Massachusetts (Remote)</i>
Adobe Research Research Intern Mentors: Yang Zhou, Zhan Xu, Jing Shi, Difan Liu, Feng Liu Project: Foundation Model for Consistent Full-Body Human Customization [C7]	May. 2024 - Aug. 2024 <i>San Jose, California</i>
Hyundai Mobis Research Collaboration Mentors: Yeong-Seok Kim, Joonhyung Park, HeonJeong Chu Project: Correspondence-aware Exemplar-based Semantic Image Synthesis [C8]	Apr. 2024 - Dec. 2024 <i>Seoul, Korea (Remote)</i>
Naver Cloud Research Intern Mentors: Seunggyu Chang, Heesu Kim, DongJae Lee Project: Training-free Semantically Consistent Image Generation [C4]	Apr. 2023 - Oct. 2023 <i>Seoul, Korea</i>

PUBLICATION

(P: Preprint, C: Conference, J: Journal, U: Under Review *: Co-first, ‡: Co-corresponding)

[P3] DiTracker: Repurposing Video Diffusion Transformers for Robust Point Tracking

Soowon Son, Honggyu An, Chaehyun Kim, Hyunah Ko, **Jisu Nam**, Dahyun Chung, Siyoon Jin, Jung Yi, Jaewon Min, Junhwa Hur, Seungryong Kim
Arxiv, 2025

[P2] Deep Forcing: Training-Free Long Video Generation with Deep Sink and Participative Compression

Jung Yi, Wooseok Jang, Paul Hyunbin Cho, **Jisu Nam**, Heeji Yoon, Seungryong Kim
Arxiv, 2025

[P1] MATRIX: Mask Track Alignment for Interaction-Aware Video Generation

Siyoon Jin, Seongchan Kim, Dahyun Chung, Jaeho Lee, Hyunwook Choi, **Jisu Nam**, Jiyoung Kim, Seungryong Kim
Arxiv, 2025

[C9] Emergent Temporal Correspondences from Video Diffusion Transformers

Jisu Nam*, Soowon Son*, Dahyun Chung, Jiyoung Kim, Siyoon Jin, Junhwa Hur[‡], Seungryong Kim[‡]
Neural Information Processing Systems (NeurIPS), 2025

[C8] Appearance Matching Adapter for Exemplar-based Semantic Image Synthesis

Siyoon Jin, **Jisu Nam**, Jiyoung Kim, Dahyun Chung, Yeong-Seok Kim, Joonhyung Park, HeonJeong Chu, Seungryong Kim
International Conference on Computer Vision (ICCV), 2025

[C7] Visual Persona: Foundation Model for Full-Body Human Customization

Jisu Nam, Soowon Son, Zhan Xu, Jing Shi, Difan Liu, Feng Liu, Seungryong Kim[‡], Yang Zhou[‡]
Conference on Computer Vision Pattern Recognition (CVPR), 2025

[J1] DiffFace: Diffusion-based Face Swapping with Facial Guidance

Kihong Kim*, Yunho Kim*, Seokju Cho, Junyoung Seo, **Jisu Nam**, Kychul Lee, Seungryong Kim[‡], KwangHee Lee[‡]
Pattern Recognition (PR), 2025

[C6] MoDiTalker: Motion-Disentangled Diffusion Model for High-Fidelity Talking Head Generation

Seyeon Kim*, Siyoon Jin*, Jihye Park*, Kihong Kim, Jiyoung Kim, **Jisu Nam**, Seungryong Kim
Association for the Advancement of Artificial Intelligence (AAAI), 2025

[C5] Local All-Pair Correspondence for Point Tracking

Seokju Cho, Jiahui Huang, **Jisu Nam**, Honggyu An, Seungryong Kim[‡], Joon-Young Lee[‡]
European Conference on Computer Vision (ECCV), 2024

[C4] DreamMatcher: Appearance Matching Self-Attention for Semantically-Consistent Text-to-Image Customization

Jisu Nam, Heesu Kim, DongJae Lee, Siyoon Jin, Seungryong Kim[‡], Seunggyu Chang[‡]
Conference on Computer Vision Pattern Recognition (CVPR), 2024

[C3] Diffusion Model for Dense Matching

Jisu Nam, Gyuseong Lee, Sunwoo Kim, Hyeonsu Kim, Hyoungwon Cho, Seyeon Kim, Seungryong Kim
International Conference on Learning Representations (ICLR), 2024 **(Oral, 1.2% acceptance rate)**

[C2] Neural Matching Fields: Implicit Representation of Matching Fields for Image Matching

Sunghwan Hong, **Jisu Nam**, Seokju Cho, Susung Hong, Sangryul Jeon, Dongbo Min, Seungryong Kim
Neural Information Processing Systems (NeurIPS), 2022

[C1] Cost Aggregation with 4D Convolutional Swin Transformer for Few-Shot Segmentation
Sunghwan Hong*, Seokju Cho*, **Jisu Nam**, Stephen Lin, Seungryong Kim
European Conference on Computer Vision (ECCV), 2022

HONORS

37th Workshop on Image Processing and Image Understanding, KIBME <i>Best Paper Award</i>	2025
37th Workshop on Image Processing and Image Understanding, KIBME <i>Best Poster Presentation Award</i>	2025
Qualcomm Innovation Fellowship 2024, Qualcomm <i>Winner</i>	2024
33th Artificial Intelligence and Signal Processing, IEIE <i>Best Paper Award</i>	2023